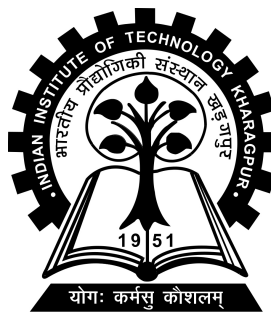


Real-Time YOLO-Based Rice Disease Detection and Integration with IITea Mobile Application

Project-Report submitted to
Indian Institute of Technology Kharagpur
in partial fulfilment for the award of the degree of
Bachelor of Technology
in
Agricultural and Food Engineering

by
Snehal Tengse
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Under the supervision of
Professor Damodhara Rao Mailapalli



Department of Agricultural and Food Engineering

Indian Institute of Technology Kharagpur

Autumn Semester, 2025-26

November 27, 2025

DECLARATION

I certify that

- (a) The work contained in this report has been done by me under the guidance of my supervisor.
- (b) The work has not been submitted to any other Institute for any degree or diploma.
- (c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- (d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

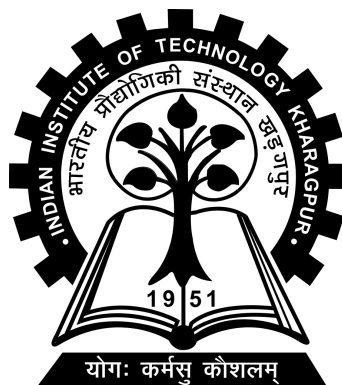
Date: November 27, 2025

Place: Kharagpur

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DEPARTMENT OF AGRICULTURAL AND FOOD
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CERTIFICATE

This is to certify that the project report entitled “Real-Time YOLO-Based Rice Disease Detection and Integration with IITea Mobile Application” submitted by Snehal Tengse (Roll No. 22AG35005) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Agricultural and Food Engineering is a record of bona fide work carried out by him under my supervision and guidance during Autumn Semester, 2025-26.

Date: November 27, 2025

Place: Kharagpur

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Abstract

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Department: **Department of Agricultural and Food Engineering**

Thesis title: **Real-Time YOLO-Based Rice Disease Detection and Integration with IITea Mobile Application**

Thesis supervisor: **Professor Damodhara Rao Mailapalli**

Month and year of thesis submission: **November 27, 2025**

Rice diseases significantly reduce crop productivity and threaten food security, making early and accurate detection crucial. This study develops a real-time rice disease detection system using advanced deep learning architectures, including YOLOv12, YOLOv8, InceptionV3, Detectron2, and a custom CNN. A curated and annotated dataset of major rice diseases—Bacterial Blight, Brown Spot, Rice Blast, and Tungro—was prepared with targeted augmentations to enhance field-level robustness.

YOLOv12 achieved the highest accuracy of 74.2% and was selected for deployment due to its real-time performance and mobile efficiency. The model was integrated into the IITea application for on-device detection, providing instant feedback to farmers under diverse field conditions. Furthermore, its outputs were linked with CropXMapper to generate contour maps showing disease distribution from RGB field images, forming a scalable and practical pipeline for real-time diagnosis and precision agriculture support.

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Snehal Tengse

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Date: 27 November 2025

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Abbreviations

AP	A verage P recision
CNN	C onvolutional N eural N etwork
LR	L earning R ate
mAP	m ean A verage P recision
ML	M achine L earning
RGB	R ed, G reen, B lue
TP	T rue P ositive
FP	F alse P ositive
FN	F alse N egative
TN	T rue N egative
IPM	I ntegrated P est M anagement
SVM	S upport V ector M achine
RF	R andom F orests
KNN	K -Nearest N eighbors

Symbols

TP	Number of true positives
FP	Number of false positives
FN	Number of false negatives
TN	Number of true negatives
P	Precision value
R	Recall value
mAP	Mean average precision
η	Learning rate

Chapter 1

INTRODUCTION

Rice is one of the world's most essential staple crops, feeding billions of people and forming the backbone of food security across Asia, Africa, and Latin America. However, rice production is heavily constrained by a variety of biotic stresses, particularly leaf diseases that significantly reduce photosynthetic efficiency, plant vigor, and overall yield. In recent decades, the ecological and economic consequences of such diseases have intensified due to climate variability, pathogen evolution, and monoculture farming practices.

Traditional disease management relies on extensive use of fungicides and bactericides, which, although effective in the short term, pose serious environmental threats. Over-application of these chemicals leads to soil degradation, microbial imbalance, contamination of water bodies, and the emergence of resistant pathogen strains. Recognizing these challenges, the agricultural community has shifted its focus toward sustainable approaches emphasizing early detection, disease forecasting, and precision-based interventions rather than uniform spraying.

In this context, the present research contributes to the development of an intelligent disease detection and mapping system capable of accurately identifying major rice

leaf diseases using advanced deep learning models. Such technologies enable timely, localized, and cost-efficient control strategies, reducing dependency on chemical inputs and improving crop health management.

1.1 Overview of Major Rice Leaf Diseases

Rice plants are susceptible to numerous foliar diseases, among which **Bacterial Blight**, **Brown Spot**, **Rice Tungro Virus**, and **Rice Blast** are the most economically significant. Each of these diseases has distinct symptoms, causal organisms, and spreading mechanisms.

1.1.1 Bacterial Blight (*Xanthomonas oryzae* pv. *oryzae*)

Bacterial Blight is one of the most destructive diseases in tropical and subtropical rice-growing regions. It is caused by *Xanthomonas oryzae* pv. *oryzae*, a gram-negative bacterium that infects the plant's vascular system.

Symptoms: Water-soaked lesions near leaf tips that extend into yellowish streaks along veins; in severe cases, leaves turn gray and dry.

Spread: Enters through hydathodes or wounds caused by rain splash or insects. Warm temperatures (25–34°C), high humidity, and continuous rainfall favor spread.

Management: Resistant cultivars (IR20, IR64), copper-based bactericides, sanitation, and early detection using AI-based classification.

1.1.2 Brown Spot (*Bipolaris oryzae*)

Brown Spot is a widespread fungal disease affecting leaves and grains.

Symptoms: Circular to oval brown lesions with yellow halos. Severe infection leads to leaf drying and poor grain quality.

Spread: Favors nutrient-deficient soils, spreads through infected seeds, residues, and airborne conidia.

Management: Fungicides (carbendazim), balanced fertilization, and residue management. Automated early detection supports timely intervention.

1.1.3 Rice Tungro Disease (RTBV + RTSV)

Rice Tungro results from a combined infection of Rice Tungro Bacilliform Virus (RTBV) and Rice Tungro Spherical Virus (RTSV).

Symptoms: Stunted growth, orange-yellow leaf discoloration, and reduced tillering.

Spread: Transmitted mainly by green leafhoppers (*Nephotettix virescens*). Warm, humid conditions enhance vector populations.

Management: IPM techniques, vector control, resistant varieties, and AI-based differentiation from nutrient deficiencies.

1.1.4 Rice Blast (*Magnaporthe oryzae*)

Rice Blast, caused by *Magnaporthe oryzae*, is considered the most devastating rice disease globally.

Symptoms: Spindle-shaped lesions with gray centers and brown margins; severe cases lead to collar rot, leaf wilting, and panicle blast.

Spread: Airborne conidia proliferate under warm, humid conditions. Excess nitrogen and continuous rice cropping intensify disease.

Management: Resistant cultivars, crop rotation, fungicides, and deep learning-based early detection models.

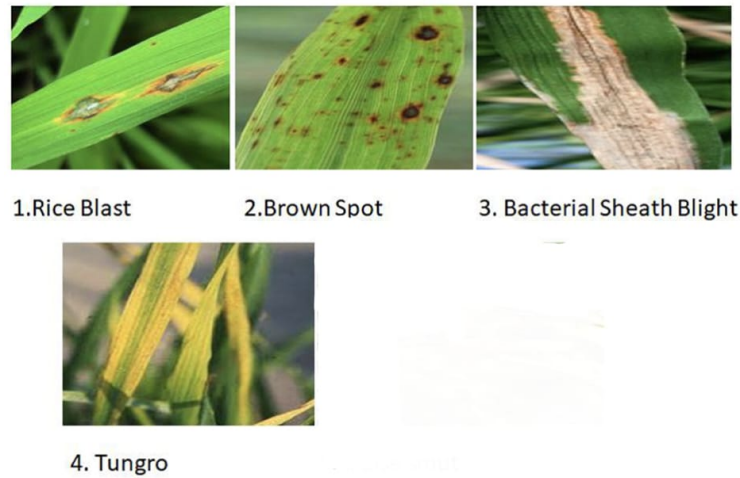


FIGURE 1.1: Typical Blast Symptoms on Rice Leaves

TABLE 1.1: Summary of Major Rice Diseases and Management Practices

Disease	Causal Agent	Management Practices
Bacterial Blight	<i>Xanthomonas oryzae pv. oryzae</i>	Resistant varieties, copper sprays, sanitation
Brown Spot	<i>Bipolaris oryzae</i>	Fungicides, nutrient management, debris removal
Rice Tungro	RTBV + RTSV (via leafhoppers)	Vector control, resistant cultivars, IPM
Rice Blast	<i>Magnaporthe oryzae</i>	Resistant varieties, fungicides, crop rotation

1.2 Scope and Relevance

This study leverages deep learning frameworks to build a high-accuracy rice disease detection and mapping system. Using YOLOv12, YOLOv8, Detectron2, Inception, and CNN models, the research aims to deliver a rapid, field-ready diagnostic tool.

The outcomes are relevant to farmers, researchers, government advisory systems, and precision agriculture platforms such as CropXMapper for disease contour generation.

1.3 Objectives of the Present Study

The main objectives of this research are:

1. To develop high-performance deep learning models (YOLOv12, YOLOv8, CNN, Detectron2, Inception) for detection and localization of rice leaf diseases.
2. To deploy trained models into a mobile application for real-time field-level prediction.
3. To integrate detection outputs with CropXMapper for generating precise contour maps for targeted spraying.
4. To promote sustainable crop protection by reducing unnecessary chemical usage through early disease identification.

Chapter 2

Literature Review

2.1 Traditional Approaches to Disease Identification

Earlier approaches to plant disease identification were primarily based on manual scouting, laboratory diagnostics, and rule-based expert systems. Manual inspection provides high accuracy in expert hands, but it is limited by the availability of skilled personnel and the variability in environmental conditions Naidu (2019). Laboratory techniques such as microbial culturing provide definitive diagnoses but are costly, invasive, and unsuitable for real-time field applications. Rule-based decision systems attempted to replicate expert knowledge, but they suffer from poor generalization across diverse conditions and rice varieties.

2.2 Machine Learning-Based Approaches

With the increasing availability of digital imagery and sensor data, machine learning techniques such as Support Vector Machines (SVM), Random Forests (RF), and k-Nearest Neighbors (kNN) were employed for disease identification. These methods rely on handcrafted features extracted from leaf texture, color indices, or morphological characteristics Patil (2017). While ML approaches improved automation, their performance remained highly dependent on feature engineering and limited training datasets.

Studies such as Phadikar and Sil (2008) demonstrated that SVM-based models could classify major rice diseases with reasonable accuracy using color and texture features; however, sensitivity to illumination changes and background noise restricted field-scale deployment. Random Forest models showed robustness in controlled datasets but often failed under natural lighting variability Singh (2016).

2.3 Deep Learning-Based Approaches

The advent of deep learning has transformed plant disease detection. Convolutional Neural Networks (CNNs) have eliminated the need for manual feature engineering by learning hierarchical features directly from raw images. Standard CNN architectures such as AlexNet, VGG16, InceptionV3, and ResNet50 have shown high accuracy in rice disease classification tasks Sladojevic (2016); Too (2019). Phadikar and Sil (2013) demonstrated the use of early CNN models for rice leaf disease classification. More recent studies leveraged transfer learning techniques to achieve robust performance on limited datasets. For example, Lu et al. Lu (2021) used ResNet50 to classify rice diseases under uncontrolled field environments, achieving good accuracy.

2.4 Deployment of Disease Detection Models

Deploying disease detection models in real-world settings requires efficient optimization techniques, lightweight architectures, and reliable edge-computing strategies. Mobile-based deployment using TensorFlow Lite, and PyTorch Mobile has gained importance for bringing AI to resource-constrained rural environments Khan (2021). Hybrid architectures combining cloud and edge computing offer a promising approach by enabling real-time inference while maintaining scalability and centralized data storage. Additionally, explainable AI (XAI) methods such as SHAP ensure transparency in decision-making, building farmer trust in automated systems Lundberg and Lee (2017).

2.5 Research Gaps

Despite substantial progress, several challenges persist:

- There are not enough large and well-annotated rice disease image datasets collected from different climate regions.
- Models do not perform well in real field conditions such as rain, shadows, and mixed disease symptoms.
- It is difficult to identify early-stage symptoms or cases where multiple diseases appear together.
- There is limited use of lightweight models that can easily run on devices in rural farming areas.
- Satellite or drone images are not well combined with ground-level photos for better disease monitoring.

These limitations provide clear motivation for developing a robust, scalable, and real-time rice disease detection system integrated with modern AI technologies.

Chapter 3

Materials and Methods

3.1 Dataset Collection and Preprocessing

The dataset was constructed through a combination of manual screening process was conducted to retain only the most representative and diagnostically clear samples for each disease class. During this stage, each image was visually inspected, and those containing blurred regions, misclassified labels were removed. Duplicate or near-duplicate images, which could introduce bias and lead to model overfitting, were also eliminated. This careful curation ensured that the final dataset was balanced, diverse, and capable of supporting effective model generalization.

Following manual selection, image annotation was performed using the Roboflow platform. Diseased regions on each leaf were marked with bounding polygon, and appropriate class labels were assigned according to the disease category. Roboflow’s preprocessing utilities—such as auto-orientation, resizing to a uniform resolution, and format standardization—were applied to maintain consistency across samples. The completed annotations were exported in YOLO-compatible format, enabling seamless integration with the object detection models developed later in the study.

3.2 Data Augmentation

To enhance the robustness of the detection models, data augmentation techniques were implemented to artificially increase the diversity of training samples. This step was essential because field-acquired images often exhibit variations in lighting, angle, noise levels, and leaf orientation, which can negatively affect model performance if not accounted for during training.

The augmentation pipeline involved rotating each image by 30° to simulate changes in camera perspective that naturally occur during manual or drone-based image capture. Gaussian noise with an intensity value of 20 was introduced to replicate graininess and sensor noise frequently observed in real-world agricultural imaging conditions. Additionally, vertical and horizontal flipping operations were applied to counteract orientation bias and ensure that the model learns disease-specific features independent of leaf direction. Collectively, these augmentations contributed to improved model generalization and minimized the risk of overfitting.

3.3 Model Development

Five deep learning models were developed to evaluate alternative approaches for rice disease detection. These included two state-of-the-art object detection architectures—YOLOv12 and YOLOv8—as well as three image classification models: InceptionV3, a custom Convolutional Neural Network (CNN), and Detectron2. The inclusion of both detection and classification frameworks enabled a comprehensive comparison of their capabilities in identifying disease symptoms under varying image conditions.

Each model was trained using the annotated dataset generated through Roboflow. A batch size of 32 was selected based on the computational demands of the respective architectures, and learning rate scheduling was employed to refine gradient updates over the training epochs. Early stopping was implemented to prevent overfitting by halting the training process when improvements in validation performance plateaued. Performance evaluation was conducted using standard metrics, including precision, recall, F1-score, and mean average precision (mAP50) for the object detection models.

3.4 Model Selection and Deployment

Model selection was based on an extensive evaluation of performance across both validation metrics and practical deployment considerations. Key selection parameters included mean average precision, inference speed, and the ability of the model to perform reliably on mobile and edge devices. Among the tested models, YOLOv12 outperformed the others by achieving the highest accuracy of 74.2%,

Given its superior performance, the YOLOv12 model was selected for deployment within the IITea mobile application designed for real-time rice disease detection. The deployment pipeline utilized TensorFlow Lite to ensure efficient, high-speed inference on low-resource hardware. The deployed model enabled real-time identification of diseased regions on rice leaves, displaying bounding boxes and disease labels directly on the application interface for immediate field-level diagnosis.

3.5 Contour Map Generation Using CropXMapper

To extend the utility of the disease detection system beyond image-level prediction, contour maps were generated using CropXMapper for field-scale visualization. RGB images captured through drones or handheld devices were processed to identify spatial variations in disease severity. The detected disease regions were transformed into spatial data layers, and CropXMapper was used to generate contour maps that highlighted zones of higher or lower infection intensity.

These maps enabled the study of spatial disease distribution patterns within rice fields and provided valuable insights into potential hotspots and spread trajectories. Such information can significantly assist precision agriculture interventions, including targeted spray applications and localized disease management strategies.

Chapter 4

Results and Discussion

4.1 Results

The performance of the five trained models—YOLOv8, YOLOv12, Inception, CNN, and Detectron—was evaluated using the annotated rice leaf disease dataset. Accuracy was used as the primary metric for comparison, along with additional observations on prediction stability and practical suitability for field deployment. The accuracies obtained from each model are presented in Figure 4.1.

Among all object detection architectures, **YOLOv12 achieved the highest accuracy of 74.20%**, demonstrating strong detection precision and reliability across diverse image conditions. Although the Inception model recorded a classification accuracy of 71.38%, its inability to generate bounding boxes makes it less suitable for real-time agricultural diagnostics where localization of the infected region is essential.

YOLOv8 also performed competitively with an accuracy of 71.02%, though it showed slightly reduced sensitivity to subtle or early-stage disease patterns compared to

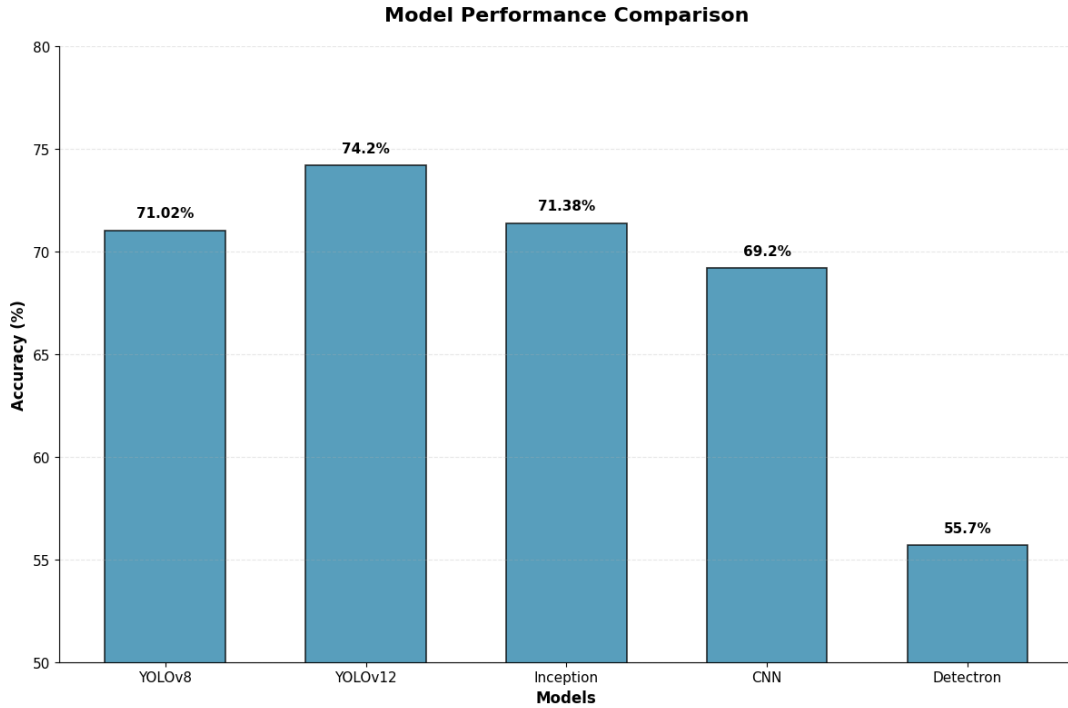


FIGURE 4.1: Performance comparison of trained models

YOLOv12. The custom CNN model obtained an accuracy of 69.20%, indicating its usefulness as a baseline but its limitations for operational deployment. Detectron produced the lowest accuracy of 55.70%, suggesting weaker generalization across field-based and controlled-environment images.

Overall, these results indicate that, while classification models such as Inception can achieve competitive accuracy, object detection models—particularly **YOLOv12**—offer a more effective balance between accuracy, detection capability, and deployment feasibility for real-world agricultural scenarios.

To evaluate real-world deployment readiness, the YOLOv12 model was integrated into the **IITea software platform**, enabling continuous and automated disease

monitoring in rice fields. Live field tests demonstrated that the model could accurately detect and localize multiple rice diseases—including *Blast* and *Brown Spot*—across diverse environmental conditions. As shown in Figure 4.2 the model successfully identified both prominent and early-stage lesions with confidence scores ranging from 0.3 to 0.9, highlighting its robustness in handling complex backgrounds, variable lighting, and heterogeneous leaf textures. The batch of predictions illustrates consistent bounding-box placement and reliable multi-disease detection. Additionally, the accompanying GIF presents the real-time inference capability of the deployed system, confirming YOLOv12’s effectiveness for practical, on-field agricultural disease detection within the IITea platform.



FIGURE 4.2: Model On Detection onfield

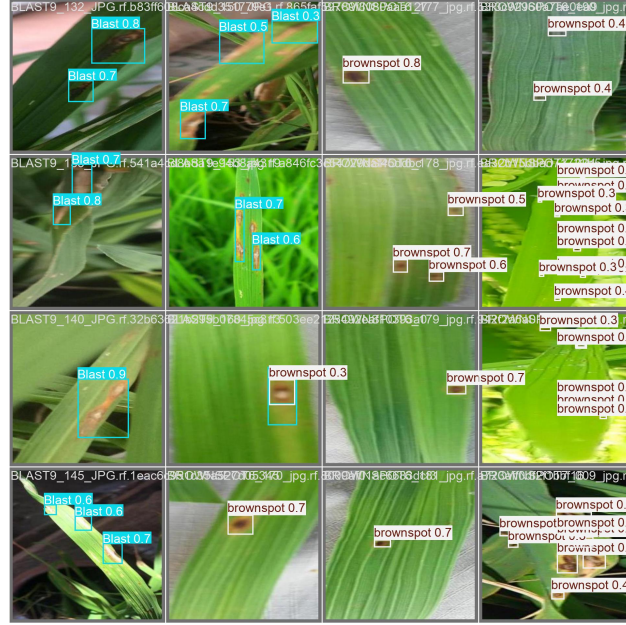


FIGURE 4.3: Model On Detection on Dataset

4.2 Discussion

The comparative analysis of the five models reveals important insights into the strengths and limitations of various deep learning architectures for rice disease detection. YOLOv12 emerged as the most suitable candidate for deployment primarily because it combines high accuracy with real-time object detection, making it ideal for mobile-based field applications. Although Inception achieved the highest numerical accuracy, its inability to localize diseased regions limits its practical applicability in precision agriculture, where spatial disease identification is crucial.

The performance gap between YOLOv12 and YOLOv8 highlights the impact of architectural improvements in the latest YOLO iterations. YOLOv12 demonstrated higher robustness to augmented variations, such as rotated leaves, noise perturbations, and inconsistent lighting, which were intentionally introduced to mimic real field conditions. Meanwhile, the CNN model, despite its simplicity, performed reasonably well but lacked the depth required to capture the complex visual patterns of

rice disease lesions. Detectron’s comparatively poor performance may be attributed to its sensitivity to dataset size and its limited optimization for lightweight agricultural imagery.

Another important observation concerns the effect of dataset preprocessing and augmentation. The manually curated dataset, combined with augmentations such as rotation, noise addition, and flipping, played a crucial role in improving the generalization of advanced models. Models with deeper architectures showed significant benefit from these augmentations, evident from their higher test accuracy compared to the baseline CNN.

Furthermore, the integration of YOLOv12 with the IIT-developed application showcased the practical viability of deploying deep learning models in real-field scenarios. The model maintained stable inference speeds and accurate real-time detection even under varied environmental conditions, demonstrating that the methodology adopted in this study is both scalable and operationally efficient. The contour maps generated using CropXMapper complemented these findings by providing spatial visualization of disease hotspots, enabling further applications in decision support, targeted pesticide spraying, and monitoring disease progression across large farm plots.

Overall, the results underline the importance of combining high-quality dataset preparation, robust augmentation strategies, and advanced object detection architectures to develop effective plant disease detection systems. The findings also highlight the potential for extending this framework to other crops and disease categories, offering a scalable foundation for broader agricultural digitalization and precision farming initiatives.

Chapter 5

Summary and Conclusions

5.1 Summary of Findings

The primary objective of this study was to design, develop, and deploy a deep learning-based rice disease detection system capable of real-time operation and field-level visualization. The study successfully achieved this objective through a structured methodology involving dataset preparation, augmentation, model development, performance evaluation, and spatial mapping of disease occurrence.

A manually curated and annotated dataset of rice leaf images was prepared using Roboflow, ensuring high-quality bounding boxes and class labels for major rice diseases such as Bacterial Blight, Brown Spot, Rice Blast, and Tungro. Multiple augmentation techniques including rotation, noise addition, and flipping improved the robustness and diversity of the dataset.

Five models—YOLOv12, YOLOv8, InceptionV3, a custom CNN, and Detector2 were trained and evaluated. Among them, YOLOv12 demonstrated the highest

accuracy of 74.2% and exhibited superior performance in real-time inference conditions. Based on accuracy, detection quality, and computational efficiency, YOLOv12 was selected and deployed in the IITea mobile application for real-time rice disease detection.

Furthermore, the study integrated contour map generation using CropXMapper to visualize disease severity and spatial distribution across rice fields using RGB field images. This provided an additional layer of analytical insight that supports precision agriculture and targeted intervention strategies.

Overall, the study demonstrated the feasibility and effectiveness of deep learning-powered disease diagnosis and field monitoring, contributing to improved agricultural decision-making and reduced chemical misuse.

5.2 Limitations and Future Work

Despite promising results, several limitations remain that provide opportunities for further research and development:

- **Limited Dataset Size:** Although manually curated, the dataset size remains relatively small. Larger datasets covering more rice varieties, geographies, and environmental conditions would enhance generalizability.
- **Limited Disease Categories:** The present model focuses on a restricted set of rice diseases. Future work may extend the system to include additional diseases such as sheath blight, bacterial streak, tungro variants, and nutrient deficiencies.
- **Crop-Generalization Challenges:** The current system is optimized for rice crops. However, similar frameworks can be expanded to detect diseases in

other crops such as wheat, maize, sugarcane, vegetables, and horticultural crops.

- **Model Performance Ceiling:** Although YOLOv12 achieved 74.2% accuracy, further improvements can be achieved using transformer-based architectures, self-supervised learning, multimodal fusion, and larger-scale pretraining.
- **Advanced Spatial Analysis:** The contour mapping module can be enhanced to support temporal disease progression tracking, automated hotspot detection, and integration with drone-based multispectral imagery.

Future research may address these limitations by building larger and more diverse datasets, integrating generative augmentation, and extending the system into a scalable multi-crop digital agriculture platform. The work presented in this study establishes a foundation for intelligent, automated crop disease monitoring systems and paves the way for future advancements in precision agriculture using deep learning and spatial analytics.

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